

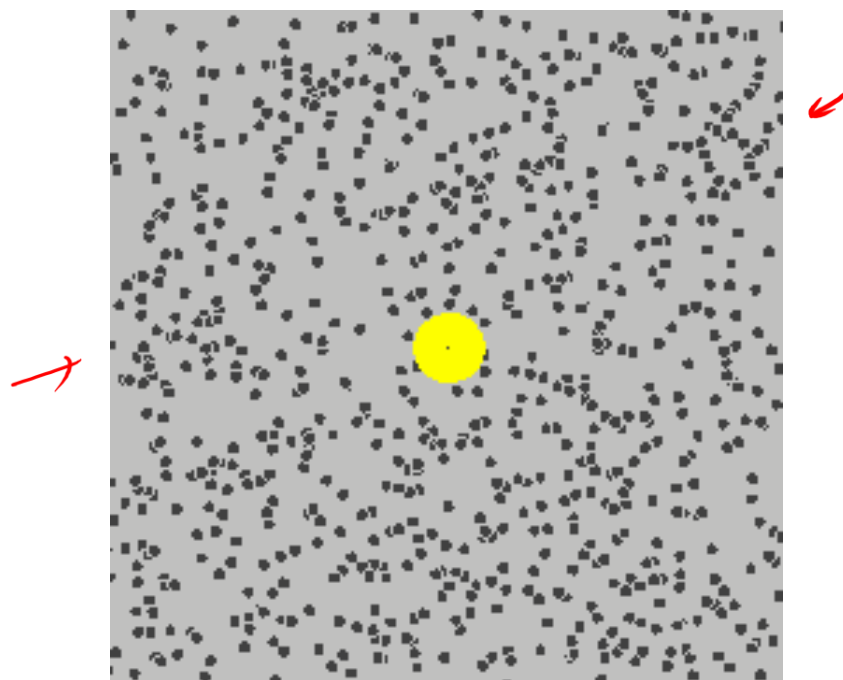
Stochastics

(2. The Brownian Motion & martingales)

History of Brownian Motion



- **Brownian motion** is the random motion of particles suspended in a medium (gas or liquid). This motion is named after the Scottish botanist Robert Brown, who first described the phenomenon in 1827, while looking through a microscope at pollen of a plant immersed in water.



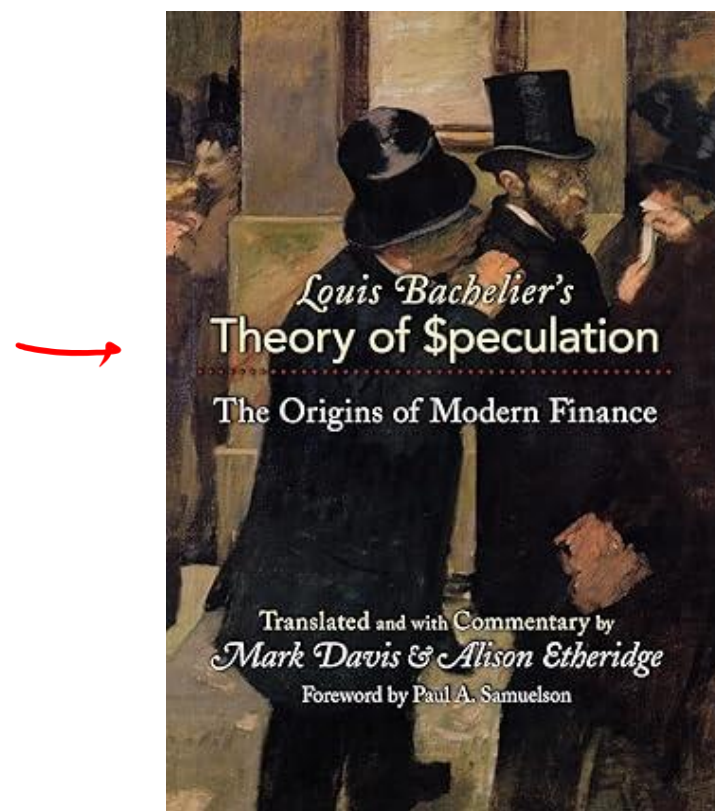
Simulation of the Brownian motion of a large particle, analogous to a dust particle, that collides with a large set of smaller particles, analogous to molecules of a gas, which move with different velocities in different random directions.

Source : Wikipedia

History of Brownian Motion



- ▶ The mathematics of much of stochastic analysis including the mathematics of Brownian motion was introduced by **Louis Bachelier** in 1900 in his PhD thesis "**The theory of speculation**", in which he presented an analysis of the stock and option markets.



The Brownian Motion (Wiener Process)

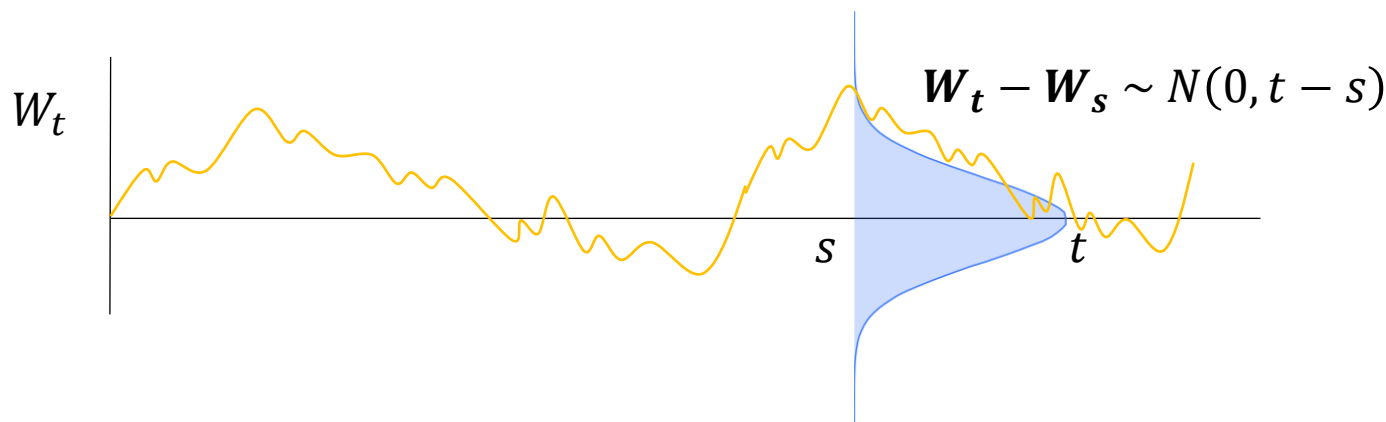
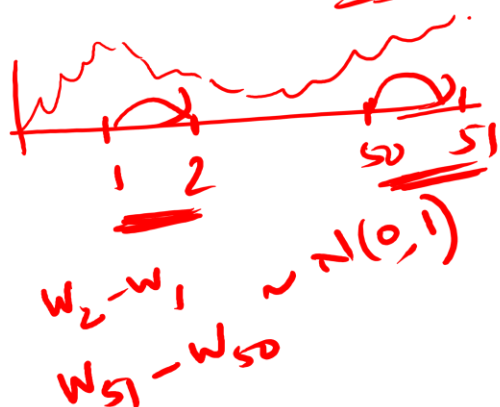
► A **standard Brownian motion** (aka Wiener process) is a stochastic process $\{W_t, t \geq 0\}$ with state space $S = \mathbb{R}$ and the following properties

B_t, X_t, S_t

□ $W_0 = 0$

□ has continuous sample path (no jumps) : $(t \rightarrow W_t)$

□ has independent stationary Gaussian Increments : $W_t - W_s \sim N(0, t - s)$



Exercise

1. Comment on the distribution of the following

- W_5

- $(W_2 - W_1)$

- $(W_{200} - W_{199}) \rightarrow$

- $2W_3 + 3$

- $\underline{\underline{dW}}$

$$\rightarrow W_5 \stackrel{0}{=} \underline{\underline{W_5 - W_0}} \sim N(0, 5)$$

$$W_2 - W_1 \sim N(0, 1)$$

$$W_{200} - W_{199} \sim N(0, 1)$$

$$\underline{\underline{2W_3 + 3}} \sim N(3, \quad)$$

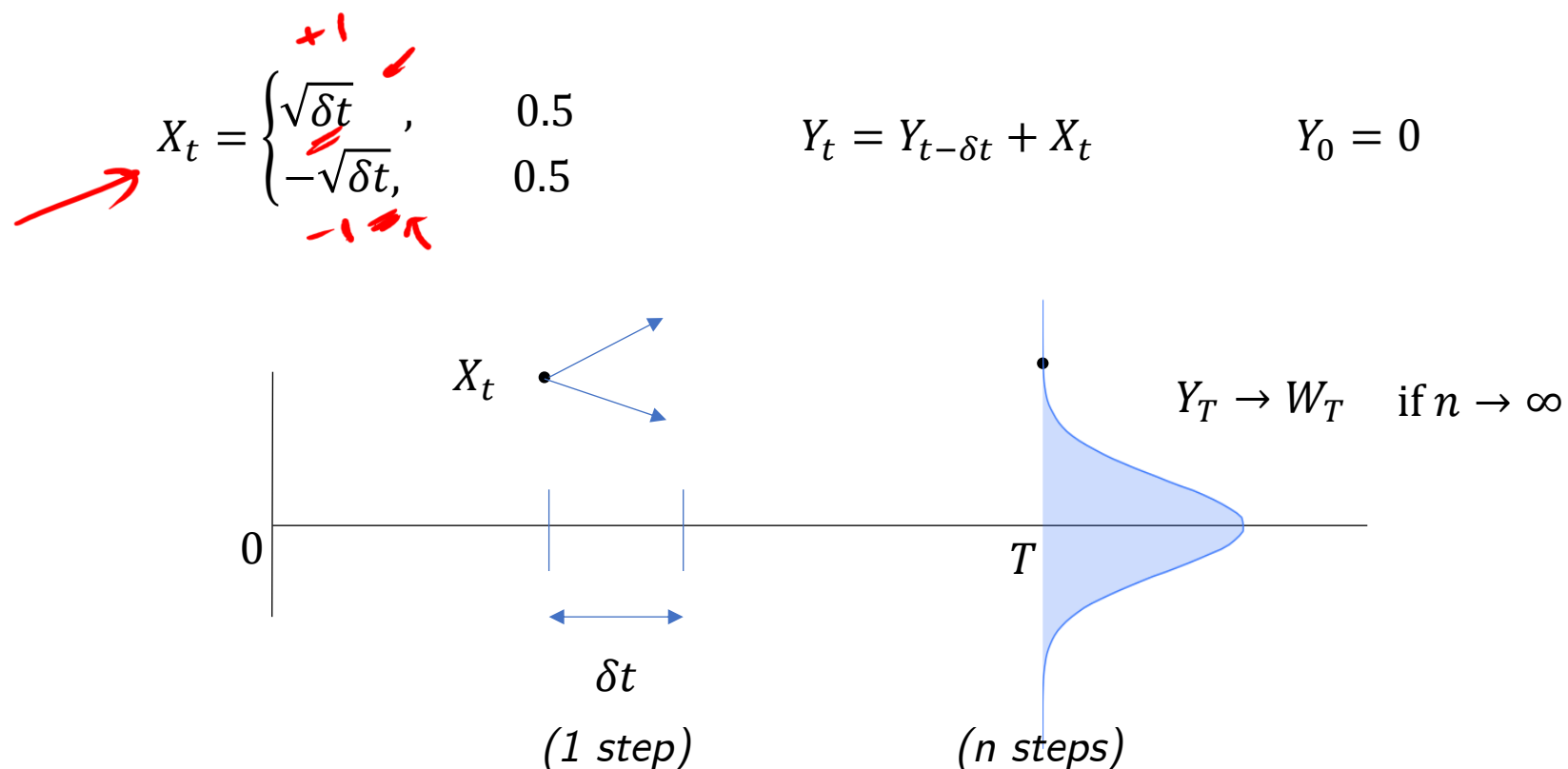
$$W_t = ? \quad W_t \sim N(0, t)$$


2. Find $\text{Cov}(\underline{\underline{W_s}}, \underline{\underline{W_t}})$ and $\text{Corr}(\underline{\underline{W_s}}, \underline{\underline{W_t}})$

3. Find $\mathbb{P}[W_1 + W_2 > 2] \rightarrow$

From Discrete to Continuous

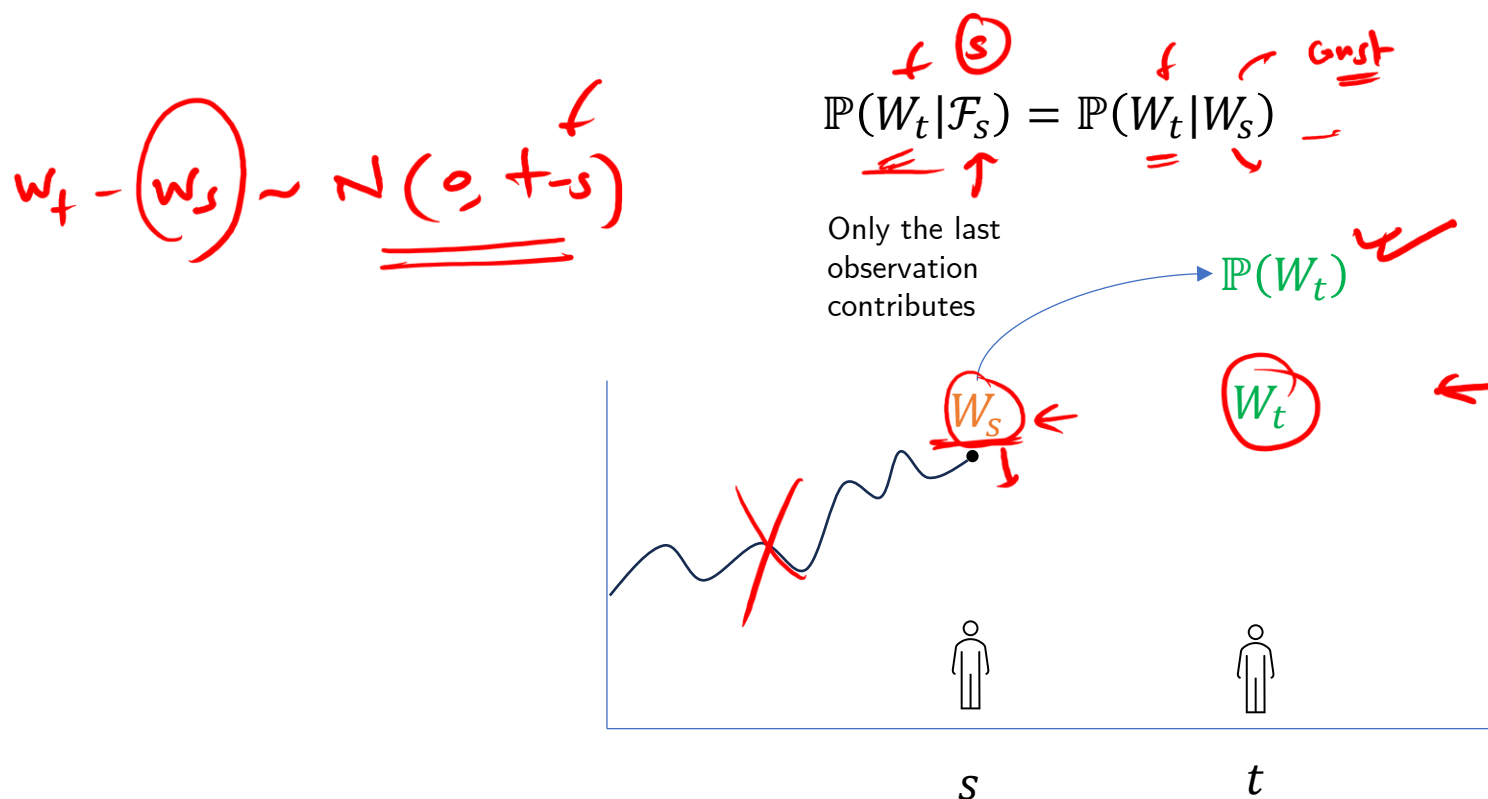
- A **standard Brownian motion** inherits several properties of simple symmetric random walk. If we use a step size of $\sqrt{\delta t}$ for a time-step δt , then the random walk is called scaled-symmetric random walk. In continuous limit, this random walk is the same as the Wiener process.



Markov



- The Wiener process is **Markov**. It follows directly from the **independent increment** property



Exercise – Given $(W_1 = 0.5, W_2 = 1, W_3 = -2)$, find the distribution of W_5 conditional on this information. What is the unconditional distribution of W_5 ?

Martingale

► A stochastic process is martingale if

→ a) $\mathbb{E}[X_t | \mathcal{F}_s] = X_s$ for all $s \leq t$

✓ b) $\mathbb{E}[|X_t|] < \infty$

✓ c) X_t is $\mathcal{F}_t$ adapted

Along with property b) and c) if

→ $\mathbb{E}[X_t | \mathcal{F}_s] \geq X_s$, it's called **sub-martingale**

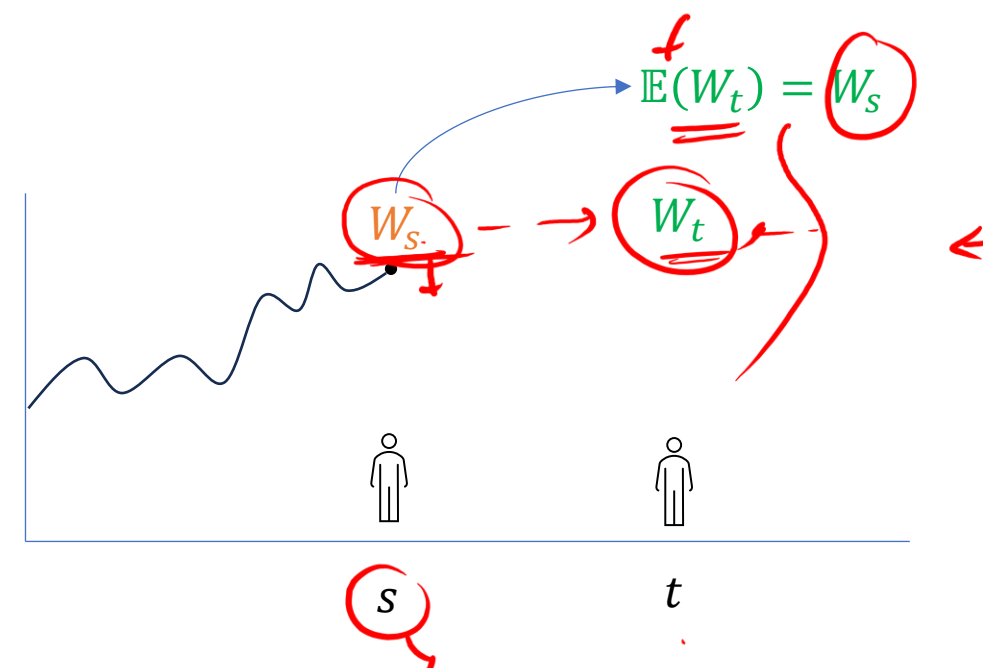
→ $\mathbb{E}[X_t | \mathcal{F}_s] \leq X_s$, it's called **super-martingale**

The Wiener process is martingale. Prove it!

Martingales are processes with no 'drift' or 'growth'.

Martingales will be the heart of derivative pricing. (More on this later!)

(Martingale Property)



How to show a process is martingale ?

► To show a process X_t is martingale, we can either show

❑ The process has constant mean $\mathbb{E}[X_t | \mathcal{F}_s] = \text{Const.}$ OR

❑ The increment has a mean of zero, $\mathbb{E}[X_t - X_s | \mathcal{F}_s] = 0.$

(the other way)

More Properties

- ▶ The Wiener Process returns infinitely many times to 0, or any other level.



- ▶ The process $W_1(t) = \frac{1}{\sqrt{c}} W_{ct}$ is also a Wiener process (standard Brownian motion). This is called the **scaling** property.

- ▶ The process $W_2(t) = t W_{1/t}$ is also a Wiener process. This is called the **inversion** property.

- ▶ Continuous everywhere, differentiable nowhere. The Brownian motion retains its squiggle or jaggedness when time step is infinitesimal. In other words, it does not become any smoother for the slope to exist.

$$dw$$

$$\left\{ \frac{dW}{dt} \right\} \text{ (does not exist)}$$

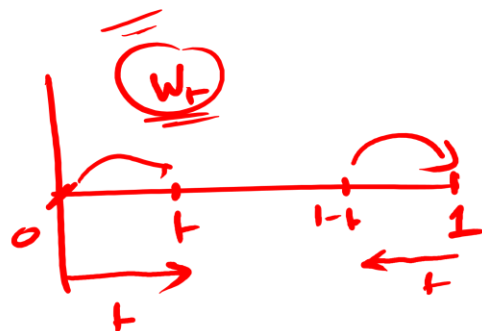
Exercise

1) **Reflection** : Show that $B_t = -W_t$ is also Wiener process.

2) **Normal Scaling** : Show that $B_t = cW_{t/c^2}$ is also Wiener process.

$$\begin{aligned}\mathbb{E}(B) &= 0 \\ \text{Var}(B) &= c^2 \times t/c^2 = t\end{aligned}$$

3) **Time Reversal** : Show that $B_t = W_1 - W_{1-t}$ is also Wiener process.



$$\mathbb{E}(B) = 0$$

$$\text{Var}(B) =$$

$$B_t \sim N(0, t)$$

Arithmetic Brownian Motion

- The Arithmetic Brownian Motion is given as

$$\sigma W_t \sim \mathcal{N}(0, \sigma^2 t)$$

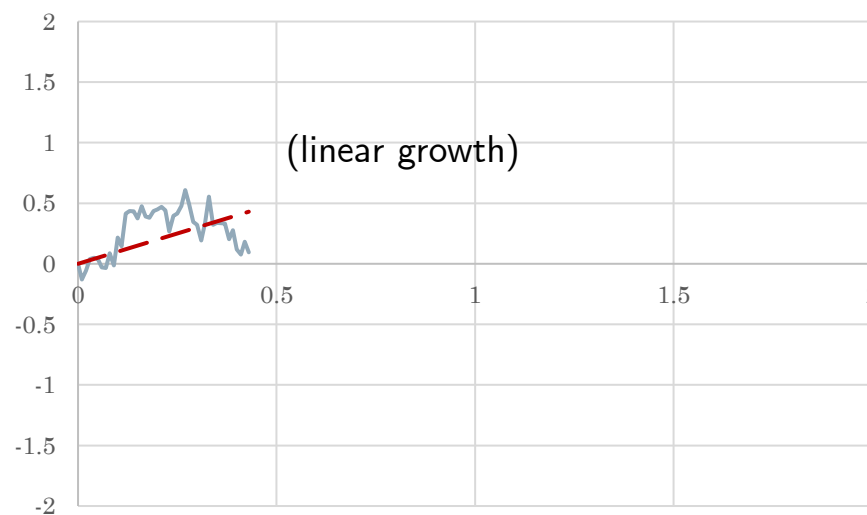
$$S_t = \underbrace{\mu t}_{\text{deterministic}} + \underbrace{\sigma W_t}_{\text{stochastic random}}$$



S_t follows **normal distribution**, with **mean** = μt and **variance** = $\sigma^2 t$.

Would you model stock price using ABM ?

$$S_t \sim \mathcal{N}(\mu t, \sigma^2 t)$$

$$\begin{aligned} \mathbb{E}(\sigma W_t) &= \sigma \mathbb{E}(W_t) = 0 \\ \text{Var}(\sigma W_t) &= \sigma^2 \times \text{Var}(W_t) = \sigma^2 t \end{aligned}$$

Geometric Brownian Motion

► The Geometric Brownian Motion is given as

$$e^x > 0$$

$$S_t = \exp(\underbrace{\mu t + \sigma W_t}_{+ve, -ve, 0})$$

$$X_t = \mu t + \sigma W_t$$

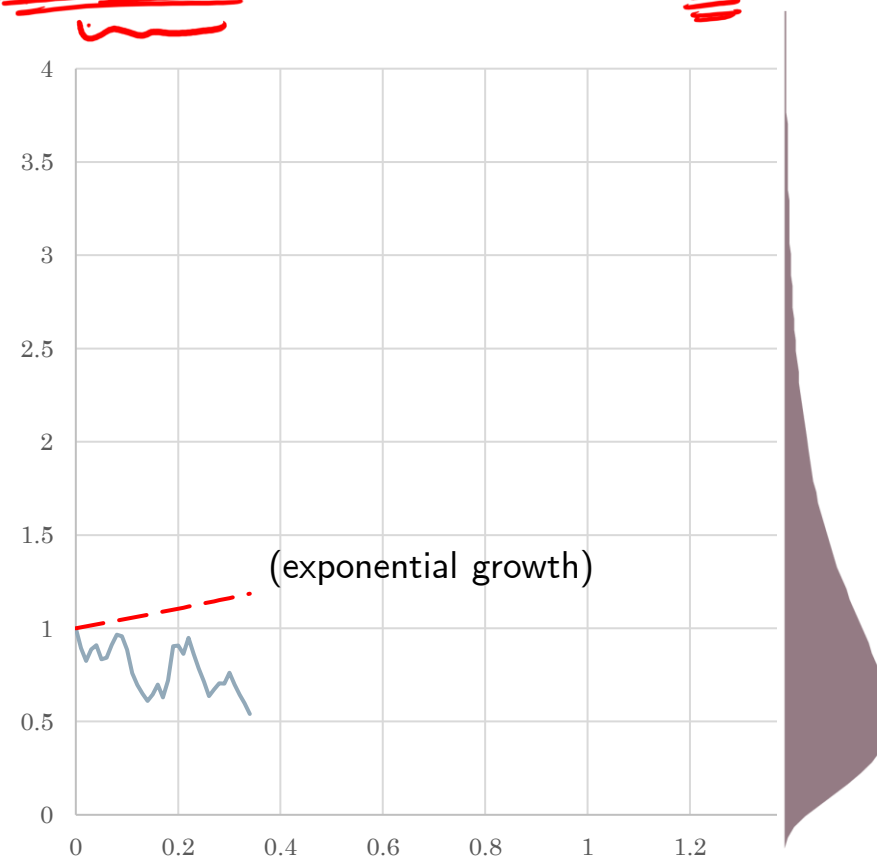
$$X_t \sim N(\mu t, \sigma^2 t), \quad S_t = e^{X_t}$$

S_t follows **lognormal distribution**, with

→ mean = $\exp\left(\mu t + \frac{1}{2}\sigma^2 t\right)$ and

variance = $[\exp(\sigma^2 t) - 1] \exp(2\mu t + \sigma^2 t)$

Alternatively, $\log(S_t) \sim N(\mu t, \sigma^2 t)$



$$E(S_t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t\right)$$

$$= e^{\dots}$$

Exercise

- 1) Let's say S_t follows ABM. Does S_t have independent increments ?
- 2) Let's say S_t follows GBM. Does S_t have independent increments ? Does $\log(S_t)$ have independent increments ?
- 3) If $X_t = \exp(\underline{W_t - W_{t-\delta}})$, find $\mathbb{E}[X_t]$

$$Y_t = \underline{W_t - W_{t-\delta}} \sim N(0, \delta)$$

$$X_t = e^{Y_t} \sim \text{lognormal}(0, \delta)$$

$$\mathbb{E}[X_t] = e^{0 + \frac{1}{2}\delta} = e^{\delta/2}$$

Problems on Martingales

- 1) A share price (S) changes every day as per the following distribution. Is S martingale ?

$$\mathbb{E}[\Delta S] = 0$$

$$10 \times \frac{1}{4} + 8 \times \frac{1}{4} - 9 \times \frac{1}{2} = 0$$

$$\Delta S = \begin{cases} 10, & \text{prob} = \frac{1}{4} \\ 8, & \text{prob} = \frac{1}{4} \\ -9, & \text{prob} = \frac{1}{2} \end{cases}$$

- 2) An asset value A_t fluctuates randomly, but there appears to be a long-term trend where the asset value increases by 2\$ per week on an average. Assuming 52 weeks per year, build a process on A_t which is martingale.

$$A_t = A_0 + (104)t + (\sigma W_t)$$

$$\mathbb{E}[A_t] = A_0 + (104)t$$

$$A_t - (104)t = A_0 + \sigma W_t$$

$$\mathbb{E}[A_t - 104t] = A_0$$

Problems on Martingales

- 3) Suppose that price increments have a continuous uniform distribution such that

$$\underline{A_t - A_s} \sim U(\underline{-(t-s)}, \underline{5(t-s)}). \text{ Construct a martingale out of } A_t$$

$$X_t \sim U(a, b)$$

- 4) Suppose $X_t = \exp\left(-\frac{1}{2}\sigma^2 t + \sigma W_t\right)$. Is X_t martingale?

- 5) Show that $(W_t^2 - t)$ is a martingale.